dynamic scaling of application instances requires stateless and disposable applications, as per the 15-Factor methodology.

This section will show you what it means for an application to be disposable, how to enable graceful shutdown, and how to scale an application in Kubernetes.

7.4.1 Ensuring disposability: Fast startup

Traditional applications deployed on application servers take quite some time to start. It's not rare for them to take several minutes before they're ready to accept connections. On the other hand, cloud native applications should be optimized for starting quickly, taking a few seconds rather than minutes to become ready. Spring Boot is already optimized for fast startup, and each new version is shipped with more improvements.

Fast startup is relevant in a cloud environment because applications are disposable and are frequently created, destroyed, and scaled. The quicker the startup, the sooner a new application instance is ready to accept connections.

Standard applications, like microservices, are good with a startup time in the range of a few seconds. On the other hand, serverless applications usually require a faster startup phase in the range of milliseconds rather than seconds. Spring Boot covers both needs, but the second use case might require a bit of additional configuration.

In chapter 16, you'll learn about serverless applications with Spring Cloud Function, and I'll show you how to package them as native images using Spring Native and GraalVM. The result will be an application with almost instant startup time, reduced resource consumption, and reduced image size.